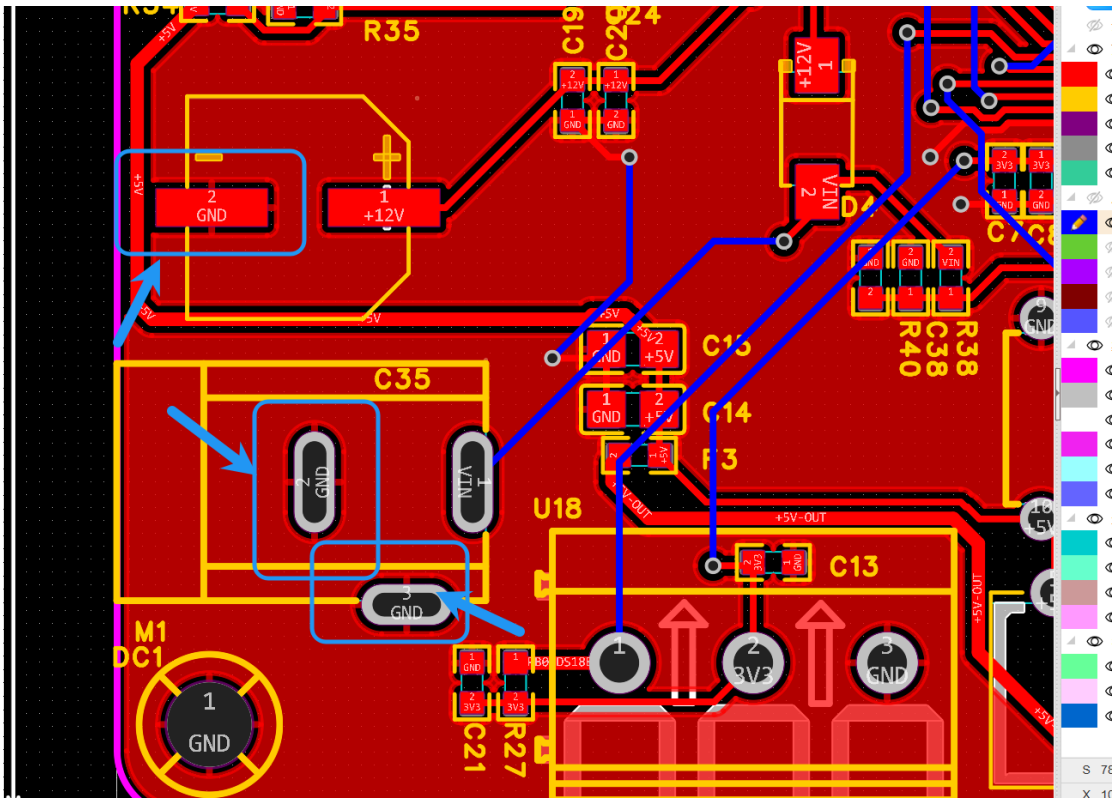
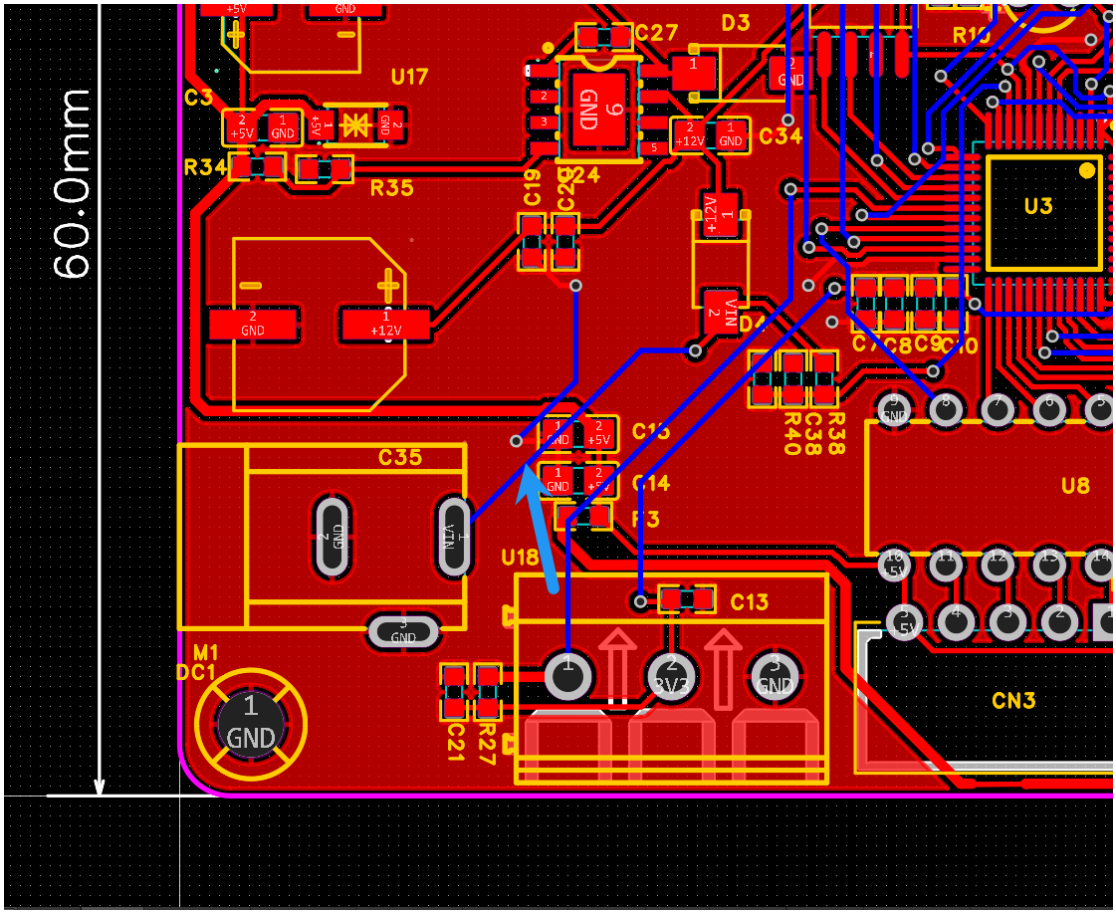
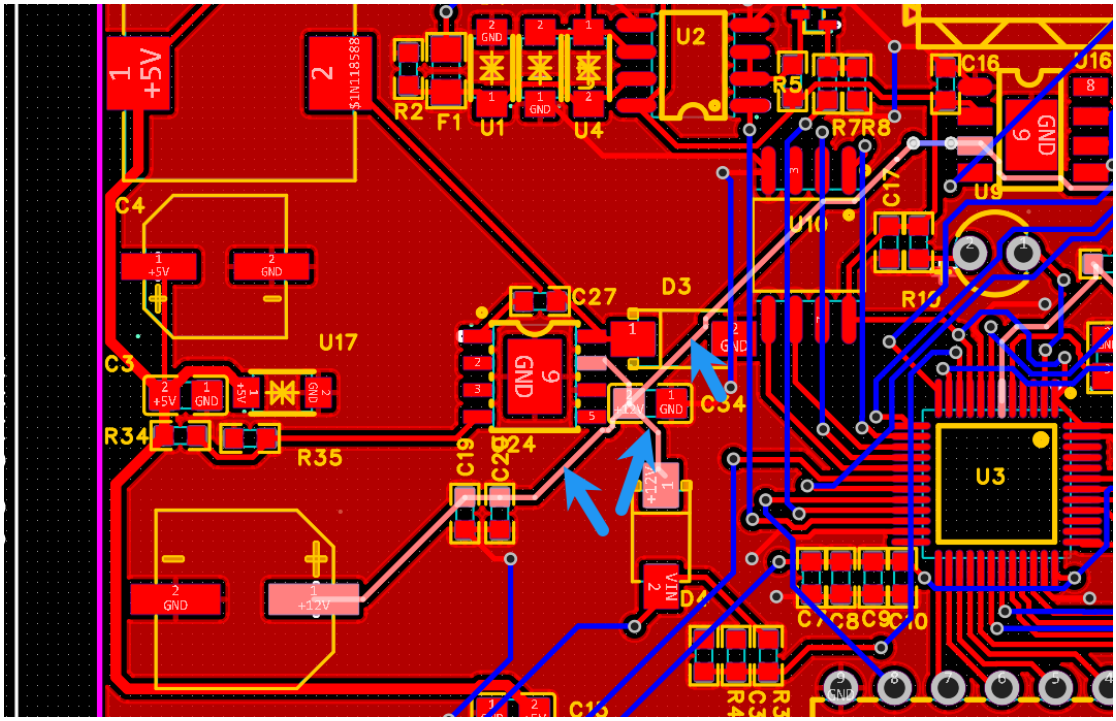
1. 一般错误：推荐使用直连铺铜，而非发散铺铜（十字铺铜）。设计→设计规则

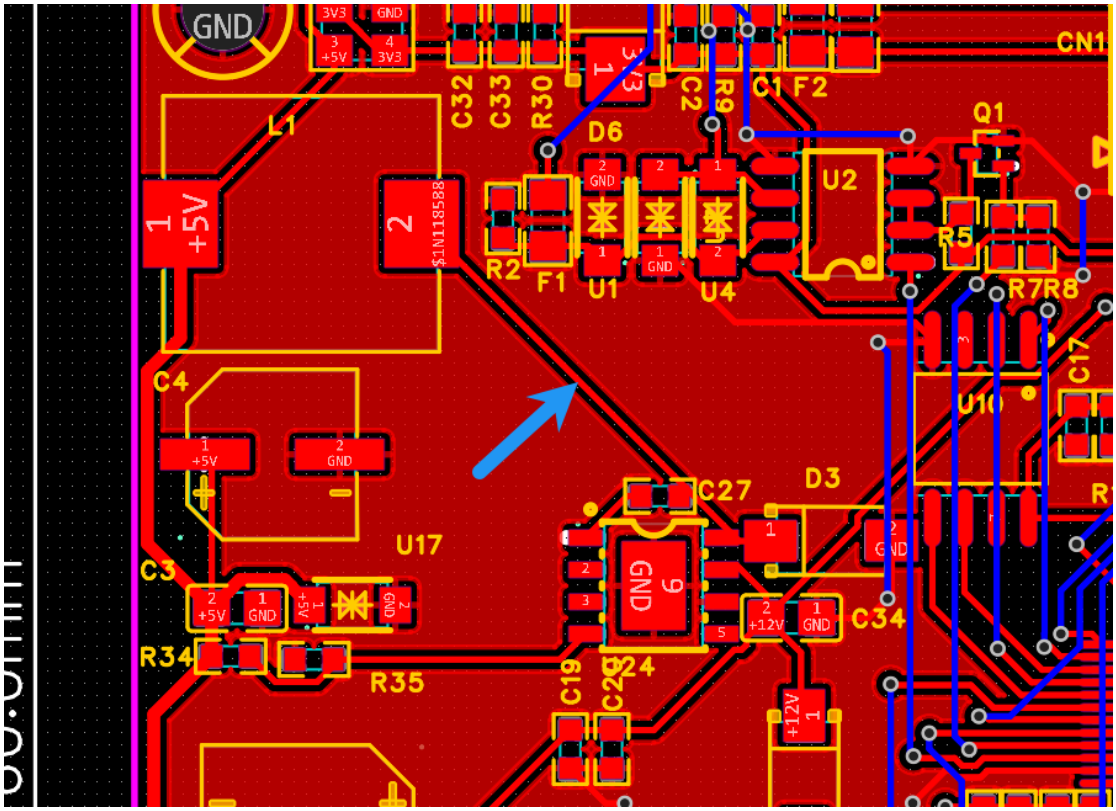


2. 严重错误：VIN 和+12V 属于供电干路，上面流过整板的电流，最好是铺铜连接，次一点需要用至少 20mil 的线宽连接，不能用信号线宽 10mil 连接

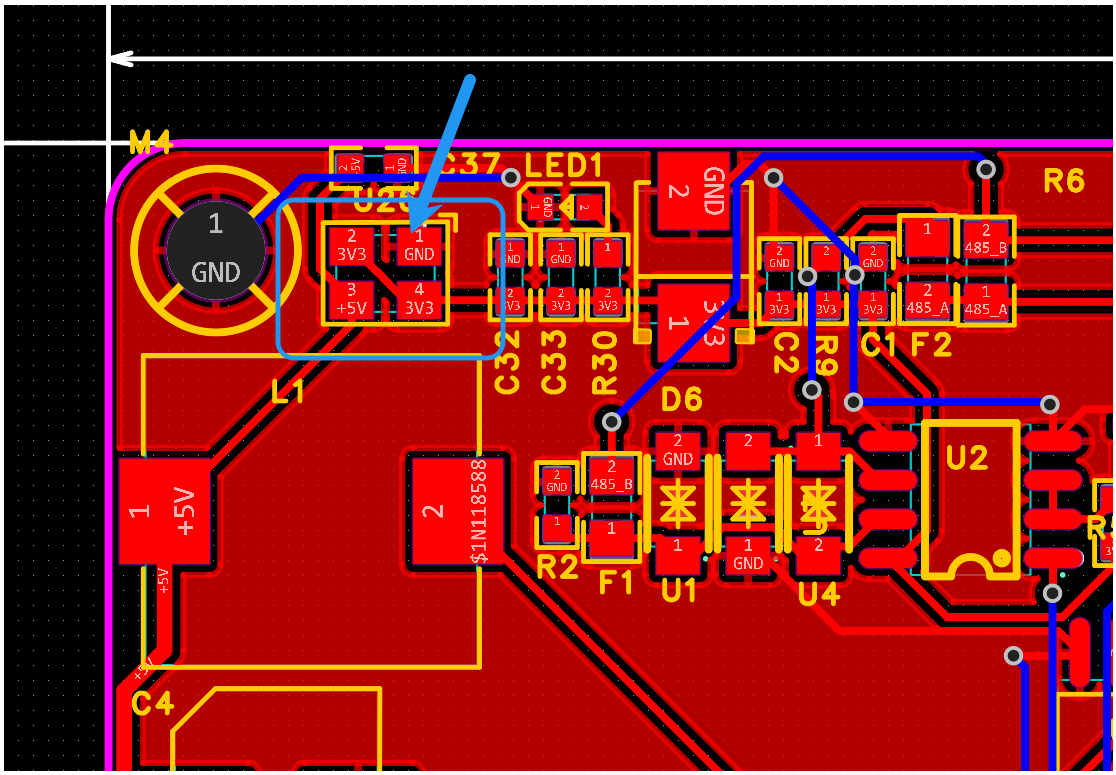




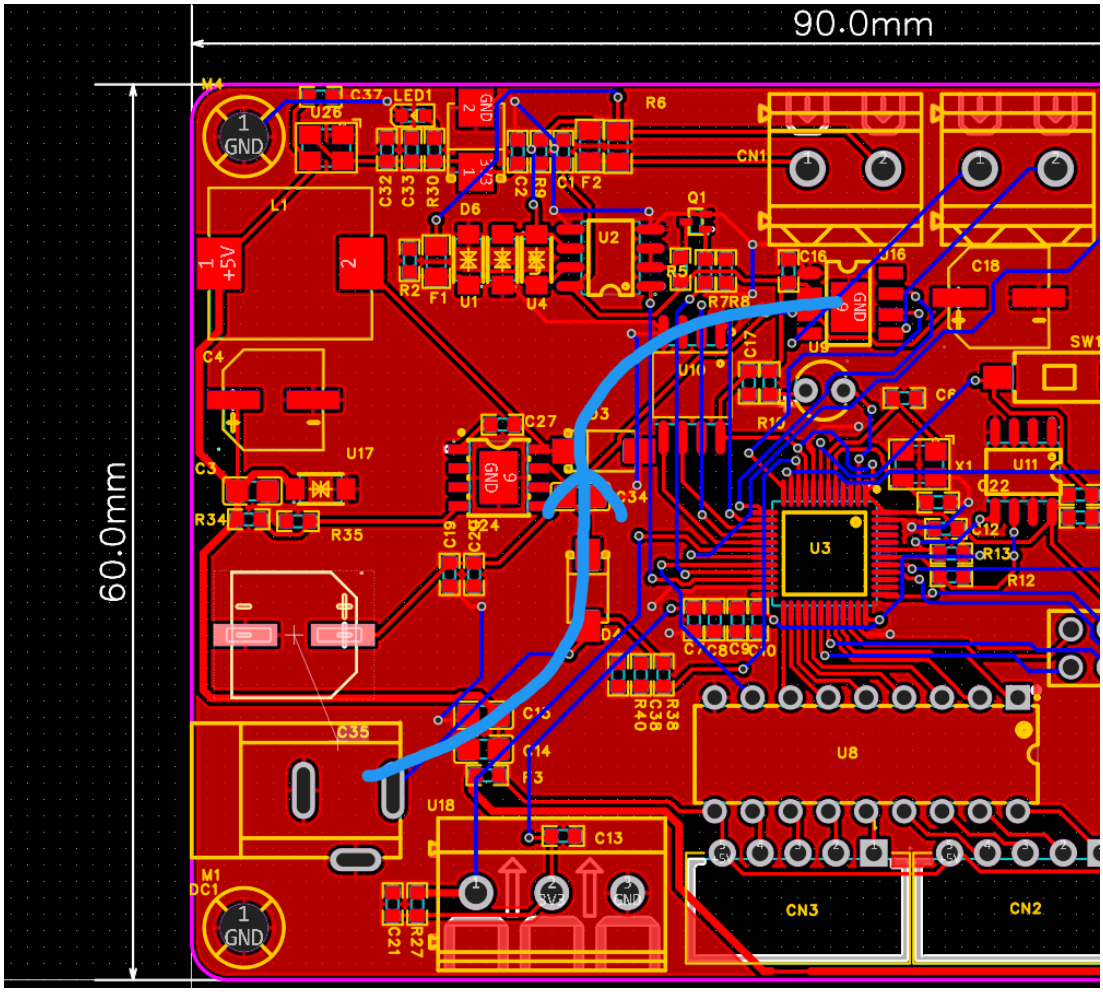
3. 严重错误：DC-DC 内部的电路连接，都是要过大电流的，也最好是铺铜连接，次一点需要用至少 20mil 的线宽连接，不能用信号线宽 10mil 连接



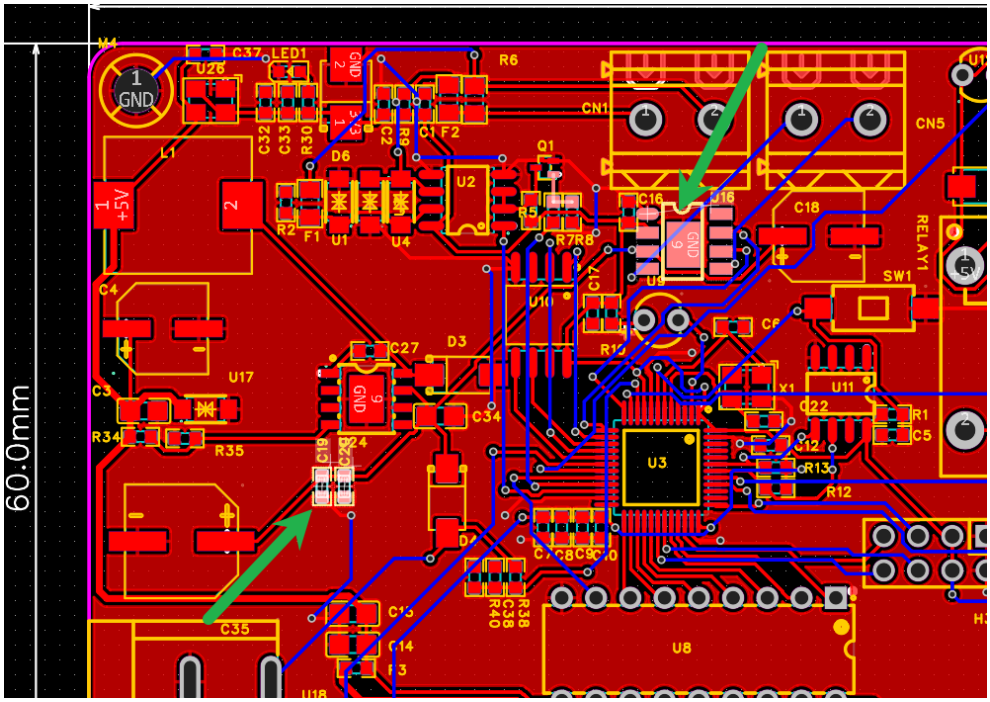
4. 致命错误：封装关联错误，此为晶振封装，不是 U26 正确的封装。

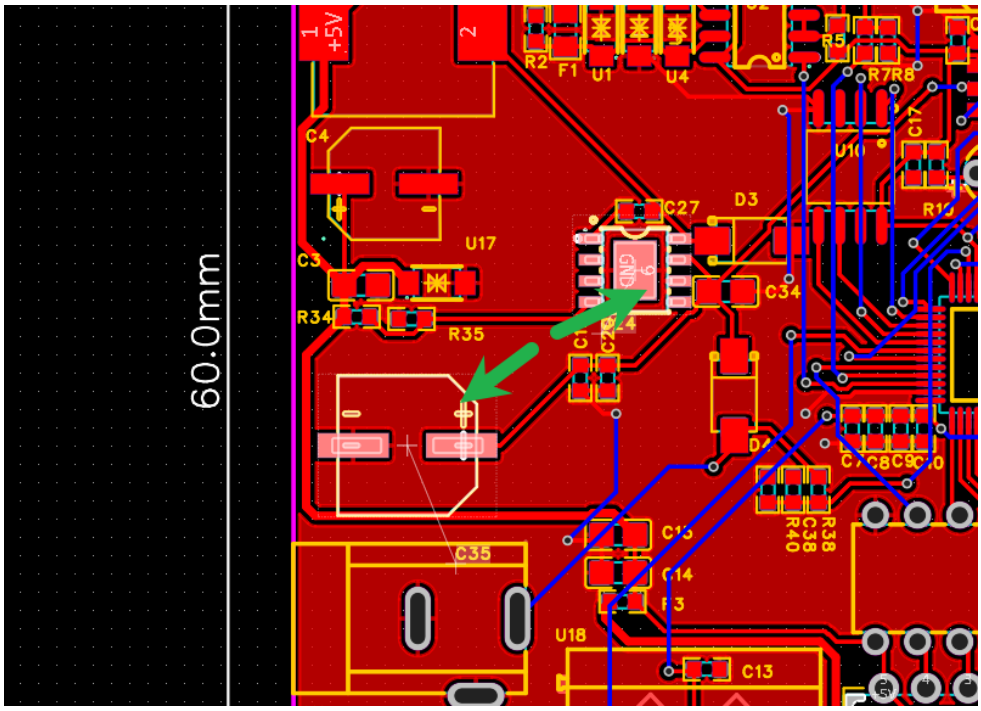


5. 严重错误：布局弯弯绕绕，12V 网络走得太远了。

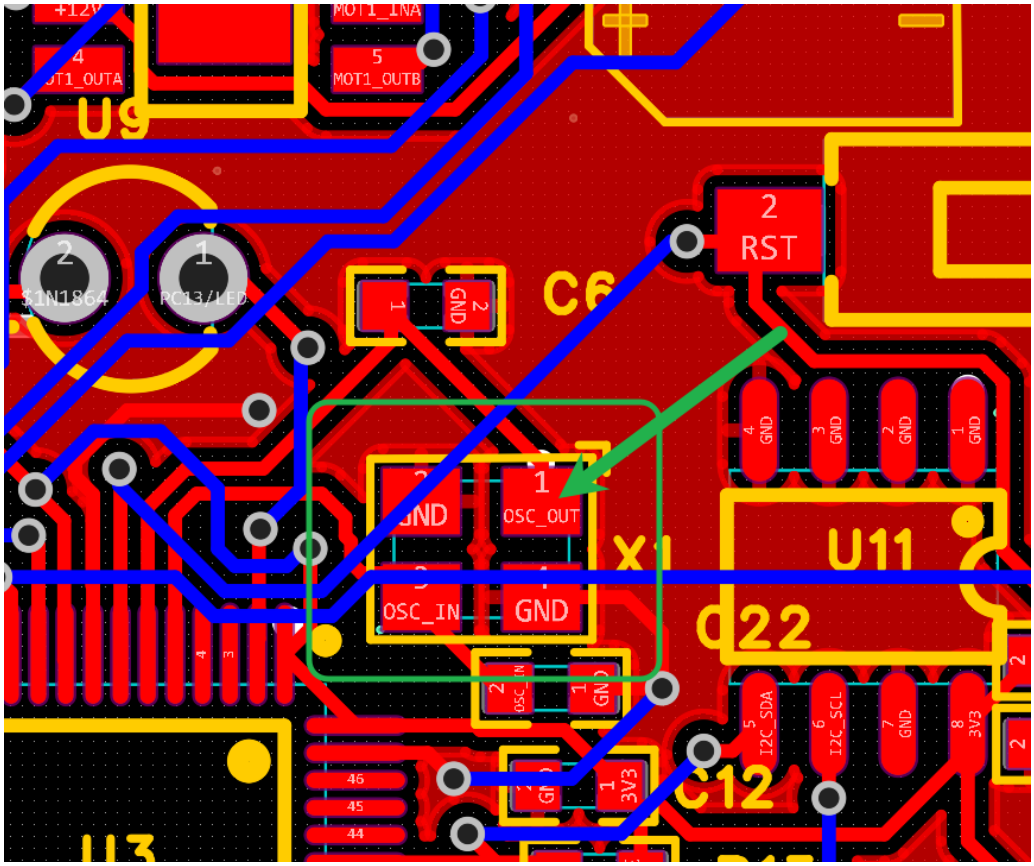


6. 一般错误：谁的电容就要靠近谁摆放。C19 和 C20 是 U16 的滤波电容，离得太远。C35 和 U24 也离得太远。

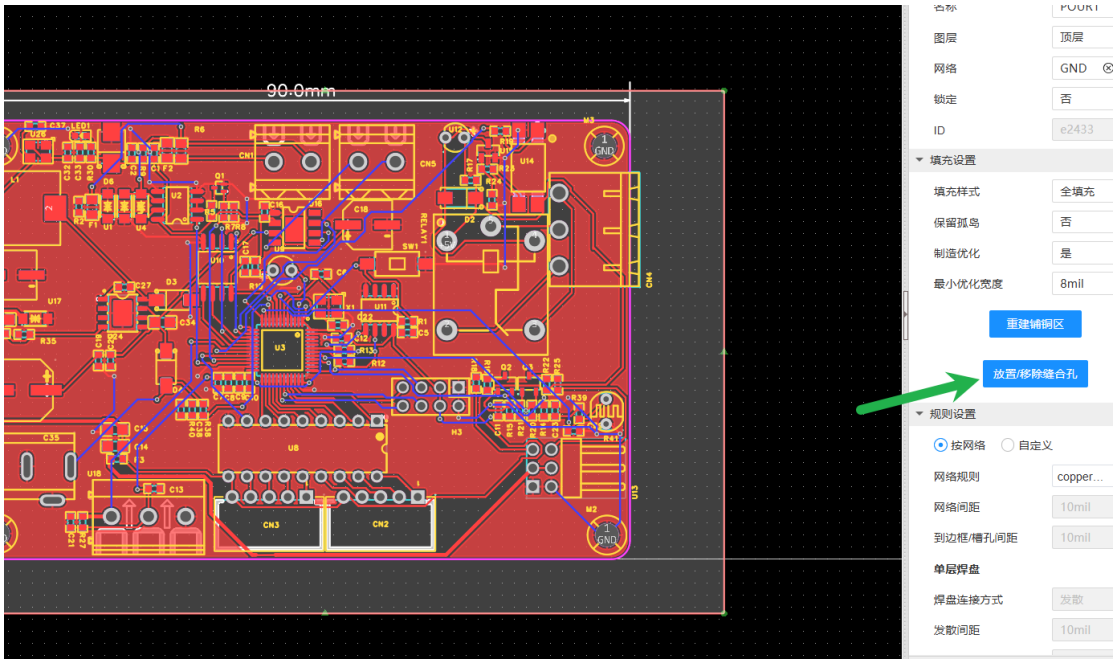




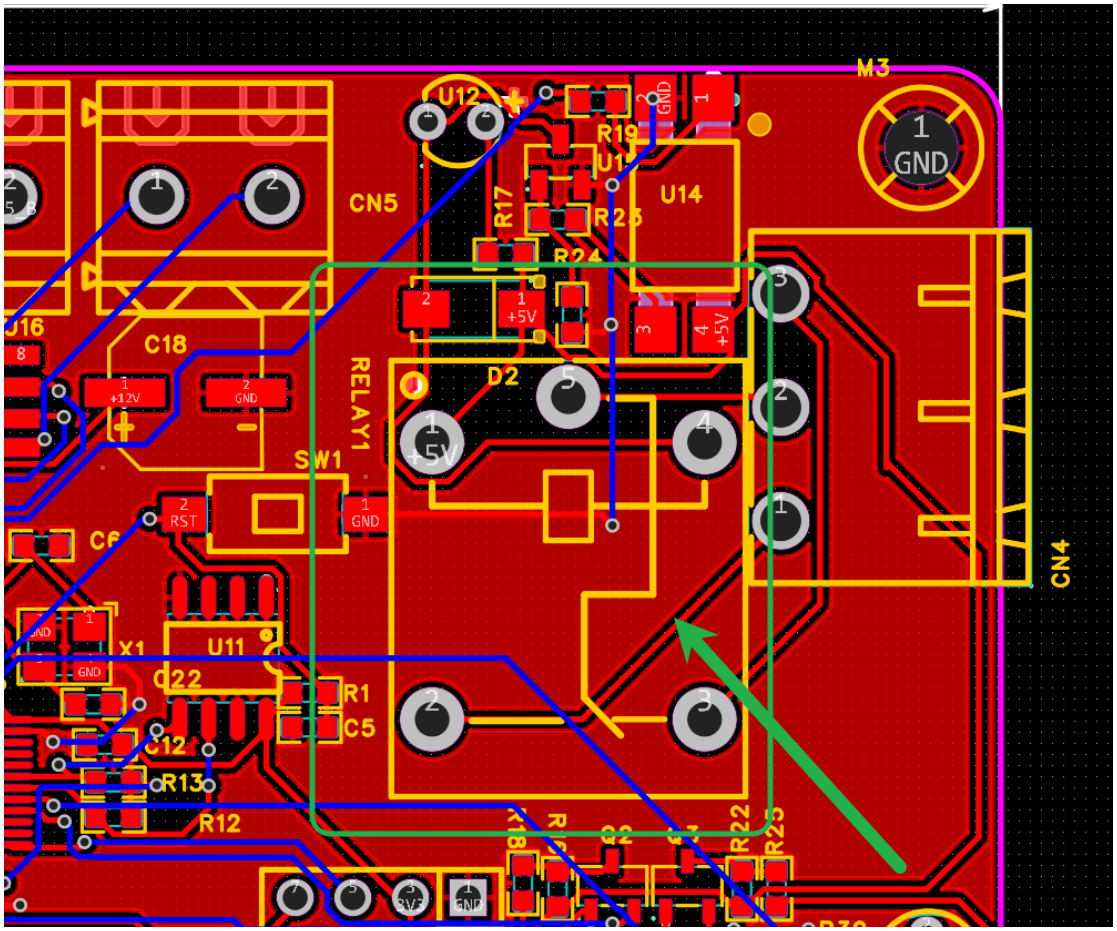
7. 严重错误：晶振需要包地和净空处理，详情参考教学视频 https://www.bilibili.com/video/BV12fcuzoELu?spm_id_from=333.788.videopod.sections&vd_source=77fc7153efddf4255fe1c7aaf03b27cb



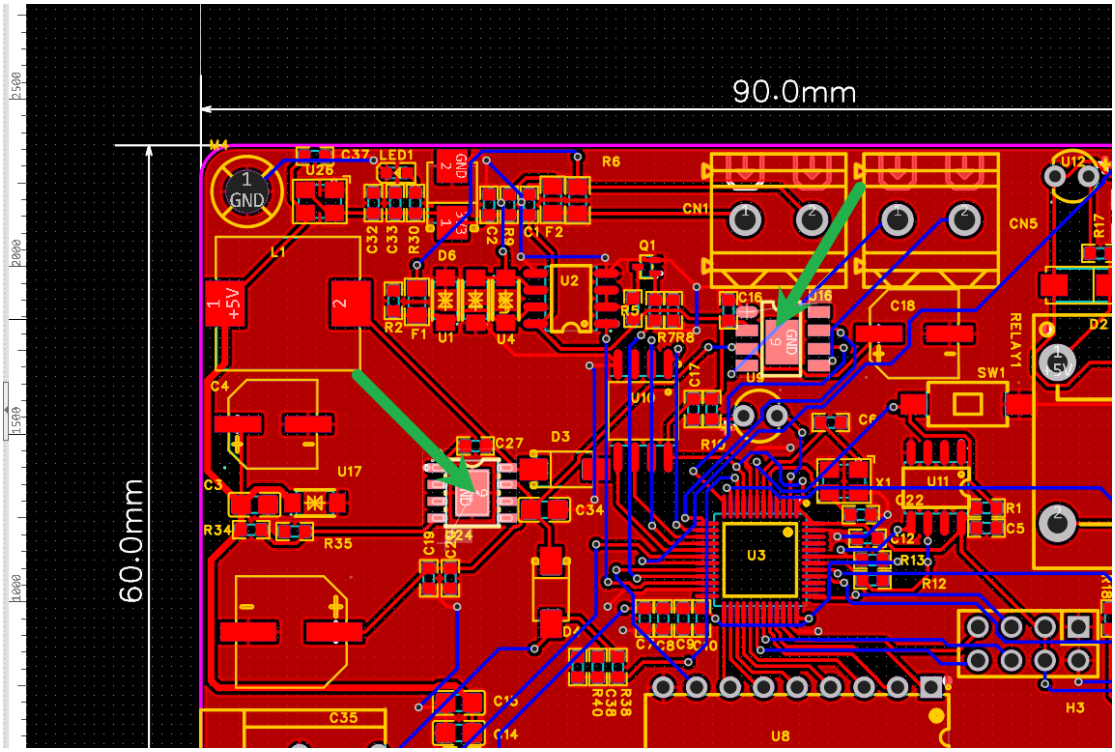
8. 一般错误：地孔不足，建议使用缝合孔功能。点击铺铜区域，右边的属性框里。



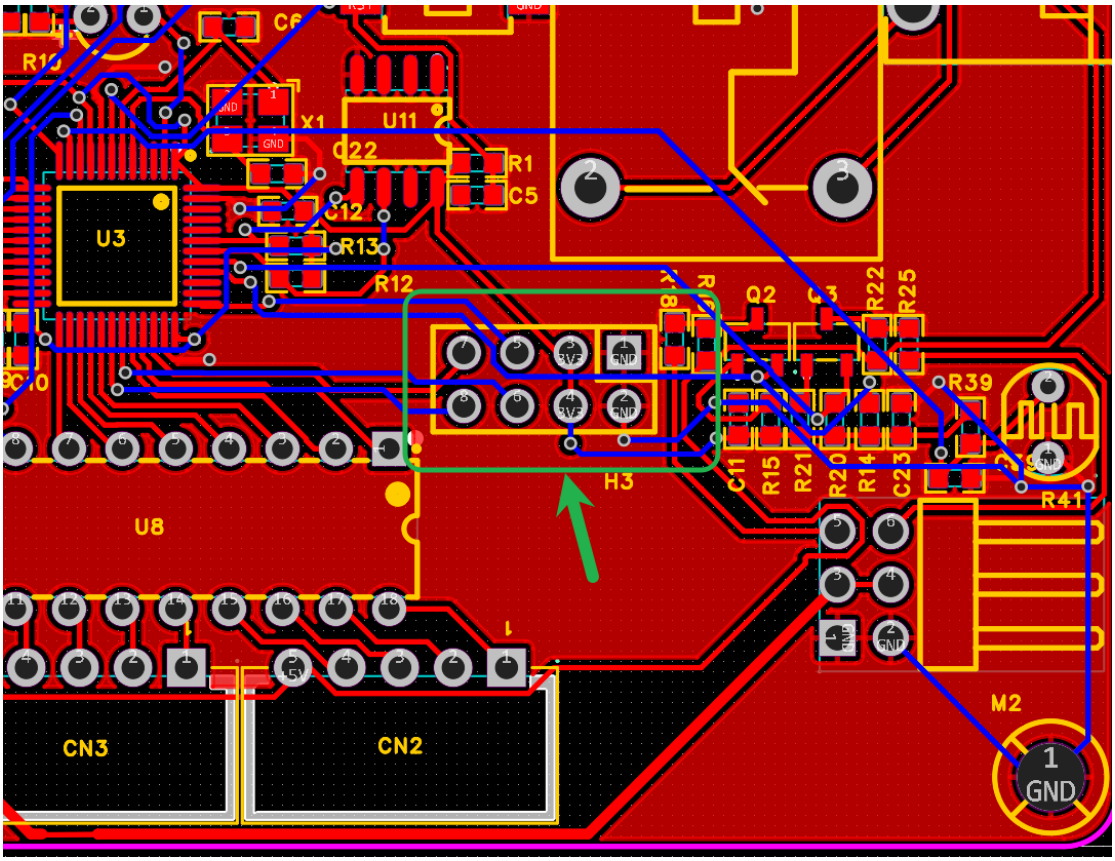
9. 一般错误：继电器所有的走线都需要加粗到至少 20mil。我们定义整板信号线 10mil，电源线至少 20mil。



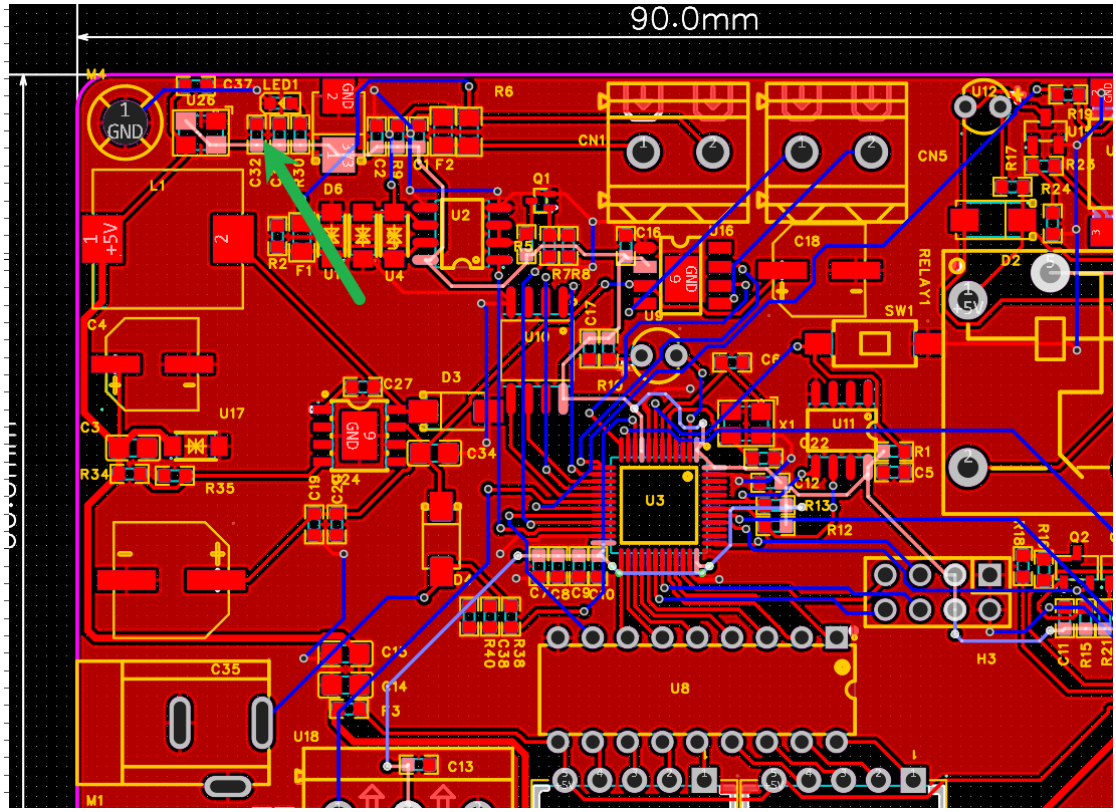
10. 一般错误：只要是电源的 GND 焊盘，都需要打过孔散热。



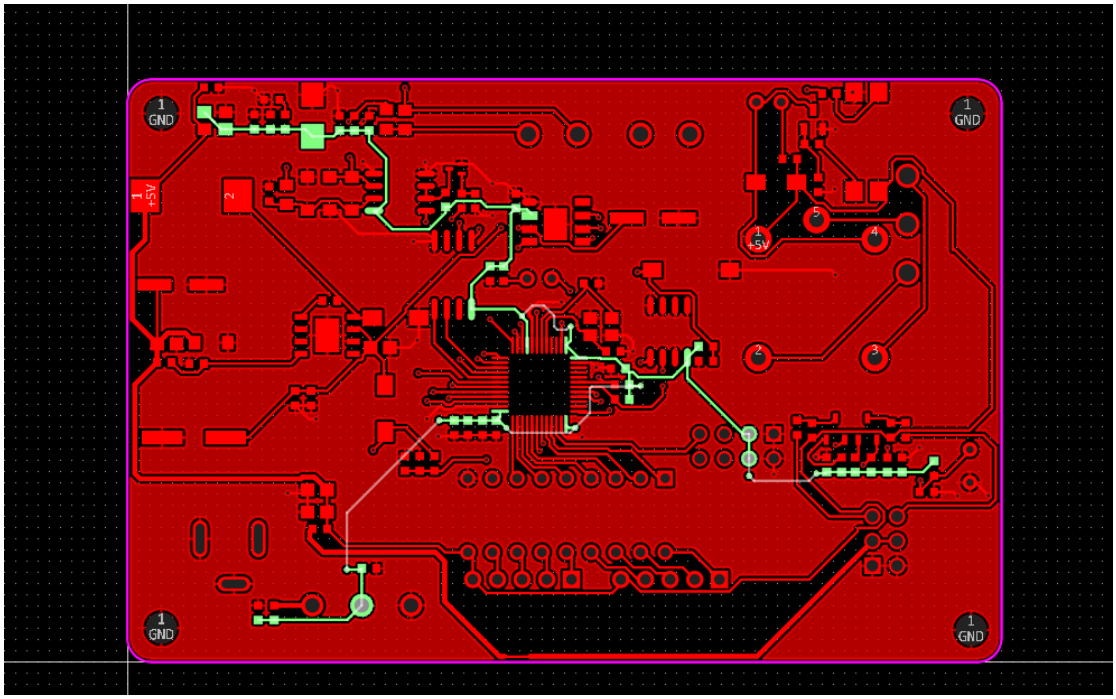
11. 轻微错误：虽然排针排母是从上往下插的，不过还是建议所有的端子往板边放置。

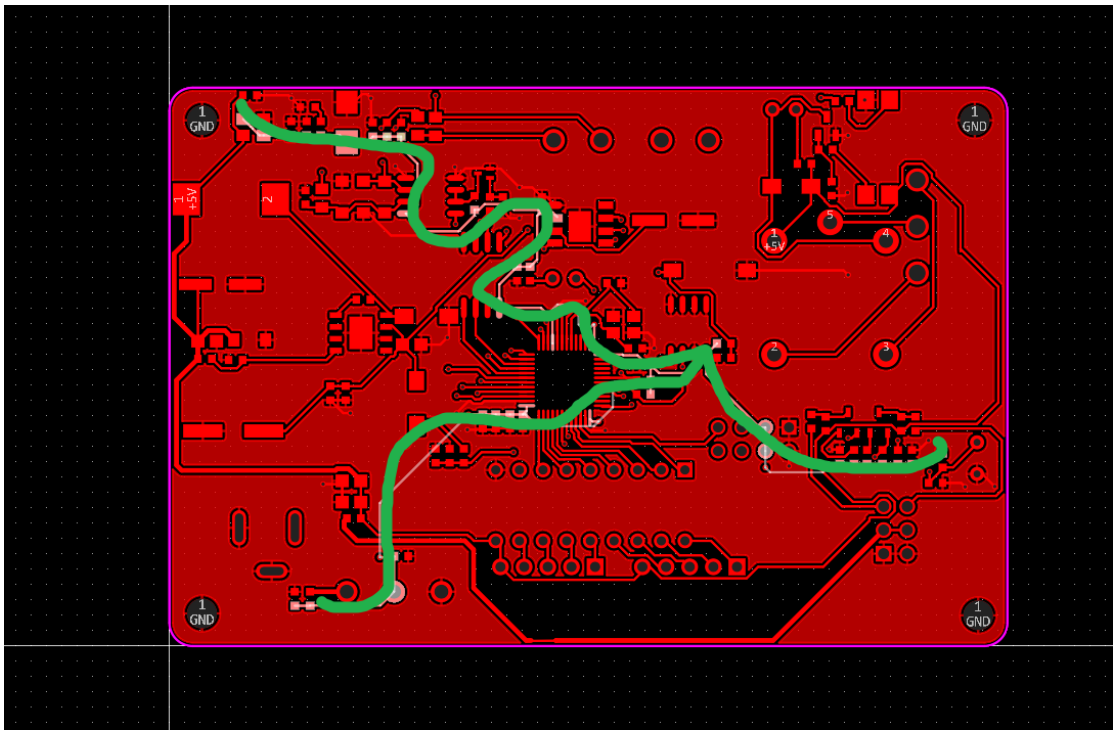


12. 一般错误：由于 3.3V 为整板供电电源，会给整板几乎所有芯片供电，所以 U26 及其外围电路不能布局在板边。

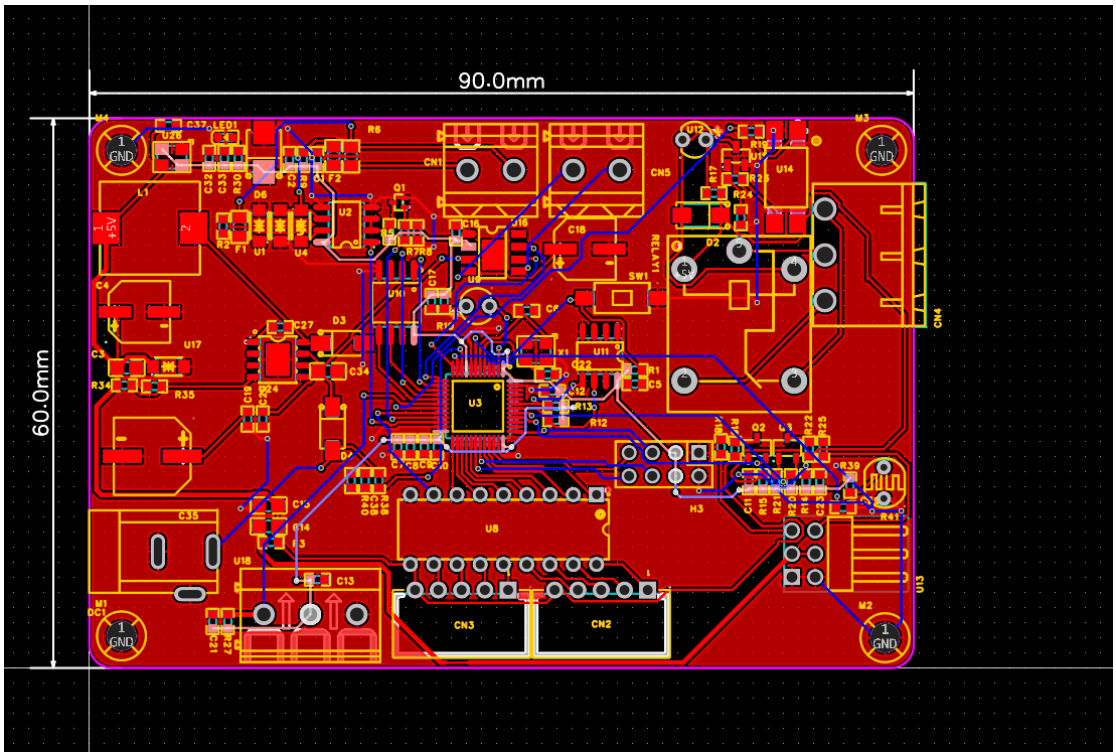


13. 一般错误：3.3V 给整板供电，需要区分干路供电和支路供电，需要确保干路至少 20mil

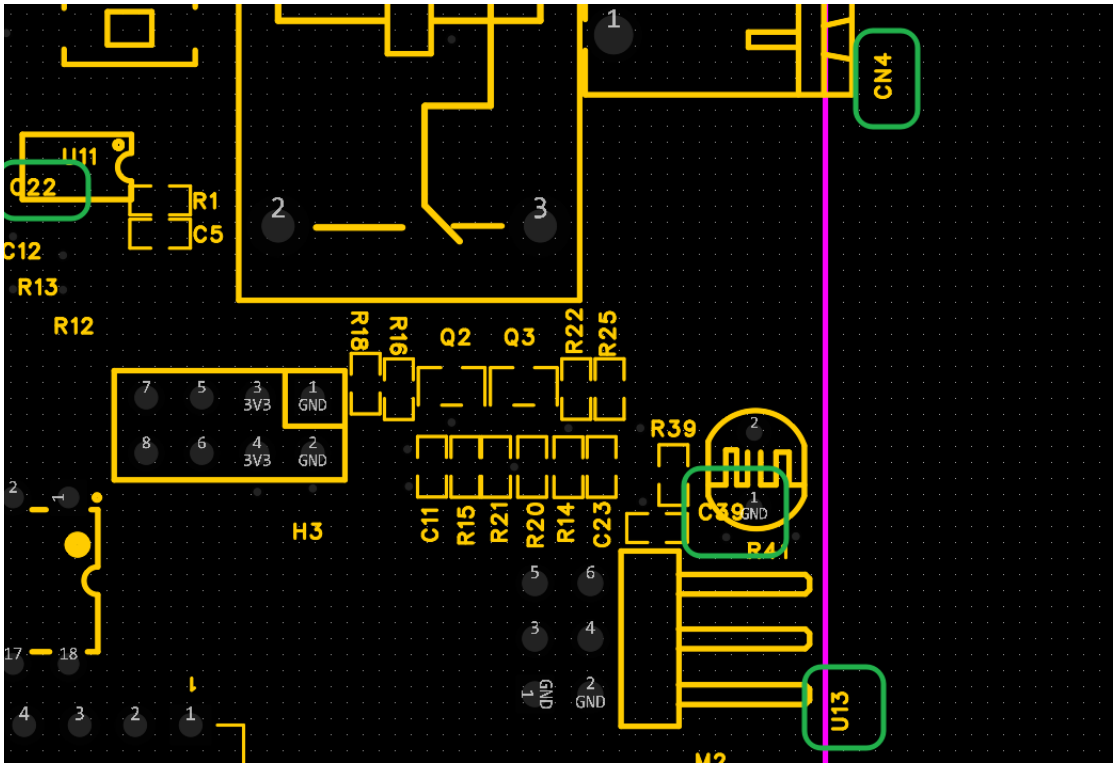




14. 严重错误：底层没有 GND 铺铜。底层的 GND 铺铜是整板主要的回流路径。



15. 轻微错误：建议走完所有走线加上泪滴，同时检查丝印，不要重叠，不要写在过孔上，不要写在板框外面，不要颠倒 180°


CN4
U11
Q22
R1
C5
C12
R13
R12
R18
R16
Q2
Q3
R22
R25
7
5
3
3V3
1
GND
8
6
4
3V3
2
GND
H3
R39
C39
GND
R4
C11
R15
R21
R20
R14
C23
5
6
3
4
1
GND
2
GND
17
18
4
3
2
1
U13